Sham Satish Thakare

Candidate for PhD in Mathematics · Discrete Mathematics, Spectral Graph Theory & Theoretical Foundations

Gold Medalist in Mathematics (2024) · Pre-Doctoral Researcher, Capital One Systems Research · APCOER Pune (SPPU)

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MATHEMATICAL RESEARCH INTERESTS

Discrete Mathematics & Graph Spectra: Spectral graph theory, hypergraph coloring, time-varying communication topologies, and combinatorial invariants in distributed state machines.
Probability & Stochastic Information Theory: Measure-theoretic convergence of stochastic sequences, analytical entropy bounds in autoregressive processes, and collinearity proofs.
Mathematical Optimization & Complexity: Convex optimization, Lagrangian duality, non-convex gradient dynamics on Riemannian manifolds, and algorithmic complexity bounds.

EDUCATION & MATHEMATICAL HONORS

APCOER Pune (Savitribai Phule Pune University), Pune, India — 2020 – 2024

Bachelor of Engineering (B.E.) in AI & DS | CGPA: 8.70 / 10.00 (First Class with Distinction) — Final Year SGPA: 9.53 / 10.00 (Rank 1)

Gold Medal in Mathematics (2024): Conferred by university leadership for highest academic standing across all mathematical and quantitative coursework in the university cohort.

Foundational Coursework Mastery: Engineering Mathematics II Tutorial (Grade O / 10.0, 100%), Discrete Mathematics (Grade A, 9/10), Advanced Linear Algebra (Spectral Decomposition, SVD), Numerical Methods, Statistics & Probability Distributions.

Verified Degree Conferral: 170/170 Credits completed, Total Grade Points: 1479/1700 (Seat No: B191102032, PRN: 72159719L, Conferred April 2024 / 125th Convocation).

Secondary Mathematics Foundations: Secondary School Certificate (SSC Class X), Happy Hours High School (March 2018, **85.20% Distinction, Mathematics: 95/100**, Seat: H006786). Higher Secondary (HSC Class XII Science, 71.23%, Seat: V001345).

Medium of Instruction: Verified on official university transcripts as entirely **ENGLISH** (Ref: EXAM./TRANS./BEN241203044).

ANALYTICAL & THEORETICAL CONTRIBUTIONS

Analytical Bounds on Token Predictive Entropy (Information Theory): Derived formal bounds for conditional Shannon entropy H(Y_t | Y_<t, X) along autoregressive reasoning trajectories; proved empirical collinearity with completion sequence length (r = +0.486, p < 10^-6), demonstrating that heuristic uncertainty-weighting acts as an implicit length penalty rather than an epistemic uncertainty measure.

Topological Quorum Intersection Invariants (Combinatorial Graph Theory): Modeled distributed dynamic quorum re-weighting as a hypergraph coloring problem over time-varying network graphs; proved that under bounded clock drift Delta and asynchronous packet arrival, dynamically re-weighted quorums strictly preserve non-empty majority intersections ($Q_i \cap Q_j \neq \emptyset$), guaranteeing state-machine linearizability.

Counterfactual Divergence Metric Formulation: Formulated the state-contrast continuation divergence D_recovery = E[Δ_recovery] - E[Δ_control], proving that isolating nominal error recovery from unconditional generative capacity requires strict covariate conditioning.

RESEARCH PREPRINTS & WORKING PAPERS

- **When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment**

  Sham Satish Thakare. Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI), 2024. [SSRN / IEEE Preprint] Formulates analytical entropy bounds across reasoning trajectories and proves sequence-length collinearity (r = +0.486, p < 10^-6) on mathematical reasoning benchmarks (GSM8K/MATH).

- **AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus**

  Sham Satish Thakare, Yuri Shelke (Distinguished Engineer, Capital One). Under Review at IEEE Transactions, 2024. Derives mathematical proofs of non-empty quorum intersection invariants ($Q_i \cap Q_j \neq \emptyset$) under bounded clock drift Delta, eliminating p99 tail

latency spikes under asymmetric partitions.

- **recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior**

  Sham Satish Thakare. Under Review at IEEE BigData, 2024. Establishes difference-in-differences estimator framework proving the necessity of counterfactual control baselines in stochastic sequence models.

RESEARCH & PRE-DOCTORAL EXPERIENCE

Capital One Systems Research, Pune, India — 06/2024 – Present

Pre-Doctoral Researcher | Mentorship: Yuri Shelke (Distinguished Engineer)

Formulated mathematical models for distributed consensus, analyzing time-varying graph topologies and proving quorum linearizability invariants.

Conducted formal statistical ablations across large-scale autoregressive models, proving that heuristic confidence weighting penalizes multi-step mathematical proofs.

APCOER Pune (Savitribai Phule Pune University), Pune, India — 06/2023 – 05/2024

Peer Academic Mentor in Discrete Mathematics & Foundations

Mentored student cohorts in rigorous mathematical deduction: set theory, modular arithmetic, graph colorings, and recurrence relations.

Organized problem-solving recitations emphasizing axiomatic proof structure and discrete optimization.

TARGET HARVARD MATHEMATICS FACULTY ALIGNMENTS

Prof. Horng-Tzer Yau: Probability theory, random matrix universality, and mathematical dynamics of high-dimensional gradient systems.

Prof. Noam Elkies: Discrete mathematics, algebraic combinatorics, lattices, and computational number theory.

Prof. Lauren Williams: Algebraic combinatorics, cluster algebras, positive Grassmannians, and discrete geometry.